PROGRAM OVERVIEW · MARCH 2026

NF Research Tools Registry Submission Forms

How researchers document, standardize, and share the tools that drive Neurofibromatosis research — and why it matters for the field.

Generated with AI assistance from Claude Code

CONTEXT

The Challenge: Fragmented Research Infrastructure

NF research spans dozens of institutions, disease manifestations, and model systems. Without a common registry, tools developed with public funding are:

- ✓ Rediscovered or rebuilt independently at each institution
- ✓ Described inconsistently across publications
- ✓ Difficult for new investigators to locate and request
- ✓ Hard for funders to track as research outputs
- ✓ Invisible to the patient and clinical communities

The goal: A single, structured, publicly accessible registry of every research tool produced by the NF community — so tools can be found, cited, and reused.

These submission forms are the front door: the standardized interface through which researchers contribute entries to that registry, hosted on Synapse.

SCALE & SCOPE

The Registry at a Glance

9	600+	150+	100+	366
TOOL CATEGORIES	CELL LINES TRACKED	ANTIBODIES CATALOGED	ANIMAL MODELS REGISTERED	TOTAL VALIDATED RESOURCES

Researcher Submits Form → **Schema Validation** → **Curation Review** → **Synapse Registry Entry** → **Community Discovery & Reuse**

Every registered tool becomes a citable, RRID-eligible record — permanently linked to the publications that describe and use it.

What Every Form Shares

All 9 submission forms — regardless of tool type — collect the same core metadata. These fields appear on every form, giving the registry a consistent backbone.

Identity

Tool name	Required
Synonyms / alternate names	Optional
Description	Required
RRID	Optional — auto-looked-up by pipeline

Provenance

Developer name	Optional
Developer affiliation	Optional
Submitter contact email	Optional

Publication Linkage — on every form

Development publication DOI	Optional — the paper that first described this tool
Usage publication DOIs	Optional — every paper that has used this tool (add as many as needed)

Acquisition — on every form

How to acquire	Required — Contact developer / Purchase from vendor / Other
→ ***Developer contact email***	*shown if "Contact developer"*
→ ***Vendor name, catalog #, URL***	*shown if "Purchase from vendor"*

PART 1

The Nine Tool Categories

Beyond the universal fields, each category captures the metadata specific to that tool type. The nine categories fall into four natural scientific groupings.

RESEARCH MODEL SYSTEMS

- Cell Lines
- Animal Models
- Patient-Derived Models
- Advanced Cellular Models

MOLECULAR REAGENTS

- Antibodies
- Genetic Reagents

ANALYTICAL & ASSESSMENT TOOLS

- Computational Tools
- Clinical Assessment Tools

INFRASTRUCTURE

- Biobanks

Research Model Systems

Systems in which NF biology is actively studied — from 2D cultures to whole animals. These forms capture how the model was established, its genetic and disease context, and how to access it.

Cell Lines

REQUIRED (BEYOND UNIVERSAL)

Species · NF disorder · Manifestation

NOTABLE OPTIONAL FIELDS

Sex, race, age · Tissue / organ of origin · STR profile / contamination status · Culture media · Doubling time · Drug resistance

TRANSLATIONAL DECISION-SUPPORT (NEW)

PK/PD capabilities · Validated drug mechanisms · Age relevance (pediatric/adult) · Timeline to results · MTA required · NGNRI repository

Animal Models

REQUIRED (BEYOND UNIVERSAL)

Species · NF disorder · Manifestation

NOTABLE OPTIONAL FIELDS

Background strain & nomenclature · Allele type · Mutation type (15+ options) · Affected gene symbol · Protein / sequence variation · Animal state, generation

TRANSLATIONAL DECISION-SUPPORT (NEW)

Induced vs. developmental model · BBB integrity status · Route of administration · PK/PD capabilities · Validated drug mechanisms · Timeline to results · Model limitations · Clinical translation history · MTA required · NGNRI repository

Patient-Derived Models

REQUIRED (BEYOND UNIVERSAL)

Model system type · Patient diagnosis · Species · NF disorder

NOTABLE OPTIONAL FIELDS

Sex, race, age at collection · Host strain (NSG, nude, etc.) · Tumor type & engraftment site · Passage number · Molecular characterization methods · Clinical data linkage

TRANSLATIONAL DECISION-SUPPORT (NEW)

BBB integrity status · Route of administration · PK/PD capabilities · Validated drug mechanisms · Clinical translation history · Timeline · limitations · MTA · NGNRI

Advanced Cellular Models

REQUIRED (BEYOND UNIVERSAL)

Model type · Derivation source · Cell types present · NF disorder

NOTABLE OPTIONAL FIELDS

NF manifestation · Matrix type (Matrigel, hydrogel…) · Culture system (bioreactor, chip…) · Culture media & supplements · Time to maturation, passage # · Characterization methods

TRANSLATIONAL DECISION-SUPPORT (NEW)

BBB model capability · PK/PD capabilities · Validated drug mechanisms · Standalone regulatory evidence flag · Timeline · limitations · MTA · NGNRI

Molecular Reagents

Physical tools used to conduct experiments — detecting proteins or manipulating genes. These forms capture the technical specifics needed for another lab to validly reproduce an experiment.

Antibodies

Antibodies used to detect NF-related proteins in experiments and diagnostics.

REQUIRED (BEYOND UNIVERSAL)

- Host organism
- Reactive species
- Target antigen
- UniProt ID

NOTABLE OPTIONAL FIELDS

- Clonality (mono / polyclonal)
- Clone ID
- Conjugate

Genetic Reagents

Plasmids, CRISPR constructs, shRNA, lentiviral vectors, and expression cassettes.

REQUIRED (BEYOND UNIVERSAL)

- Insert species
- Entrez gene ID
- Vector backbone
- Vector type

NOTABLE OPTIONAL FIELDS

- gRNA / shRNA sequence
- Cloning method & sites
- Resistance markers
- Biosafety / hazardous flag
- Insert & total construct size

> This is the most technically detailed form — capturing everything needed to order or reconstruct a plasmid from a repository like Addgene.

Analytical & Assessment Tools

Tools for making sense of data or measuring patient outcomes — these are used *after* an experiment or in the clinic, not to generate biological material.

Computational Tools

Software, pipelines, databases, and web apps built for NF data analysis.

REQUIRED (BEYOND UNIVERSAL)

- **Software type** (analysis software, pipeline, package, database, web app…)

NOTABLE OPTIONAL FIELDS

- Programming language(s)
- Version
- Source repository URL
- Documentation URL
- License type
- Container available (Docker)
- Dependencies & system requirements
- Maintainer name & organization
- Distribution channel (PyPI, CRAN, Bioconductor…)
- Analytical platform support (mass spec, metabolomics, PK/PD…)

Clinical Assessment Tools

Validated scales, questionnaires, and patient-reported outcome measures (PROMs) used in NF clinical studies and trials.

REQUIRED (BEYOND UNIVERSAL)

- **Assessment type**
- **Target population**

(pediatric / adult / disease-specific)

NOTABLE OPTIONAL FIELDS

- NF-specific vs. adapted from general use
- Number of items / questions
- Scoring method
- Psychometric validation status
- Validated languages
- Administration time
- Availability / licensing status
- Digital / electronic format available
- Cognitive/behavioral domains assessed
- Regulatory acceptance history (FDA/EMA COA endorsement)

Biobanks

Biobanks are repositories that store and distribute NF biospecimens — tumor tissue, blood, DNA, RNA, and cell preparations. Rather than a single experimental tool, a biobank entry makes an entire collection of specimens discoverable.

Biobank Form Fields

REQUIRED (BEYOND UNIVERSAL)

- Biobank URL

NOTABLE OPTIONAL FIELDS

- NF disease types covered (NF1, NF2, Schwannomatosis)
- Specimen types (tissue, blood, DNA, RNA, cells)
- Tumor types available
- Preservation methods (FFPE, cryopreserved, flash-frozen)
- Specimen formats
- Access procedure & request form URL
- Biobank website URL
- Curator / contact name & affiliation
- MTA required
- NGNRI repository affiliation

4 NF biobanks are already registered, making specimen access discoverable through a single interface for the first time.

Biobank entries require a name and URL — enough for the community to find and contact the biobank immediately. All specimen and content details can be added over time.

Required vs. Optional at a Glance

Tool Category	**Required Fields** (all forms also require tool name & description)	Key Optional Fields
Cell Lines	**Species, NF disorder, Manifestation**	Sex, race, age; tissue/organ; STR profile; culture media; doubling time; drug resistance; PK/PD capabilities; validated drug mechanisms; age relevance; timeline to results; MTA required; NGNRI repository
Animal Models	**Species, NF disorder, Manifestation**	Strain nomenclature, allele type, mutation type, gene symbol, sequence variation; induced vs. developmental; BBB integrity; route of administration; PK/PD capabilities; validated drug mechanisms; pediatric suitability; timeline to results; model limitations; clinical translation history; regulatory acceptance history; MTA required; NGNRI repository
Patient-Derived Models	**Model system type, Patient diagnosis, Species, NF disorder**	Sex, race, age; host strain; tumor type; engraftment site; characterization methods; BBB integrity; route of administration; PK/PD capabilities; validated drug mechanisms; clinical translation history; timeline; model limitations; regulatory acceptance history; MTA required; NGNRI repository
Advanced Cellular Models	**Model type, Derivation source, Cell types, NF disorder**	NF manifestation, matrix type, culture system, culture media, maturation time, QC metrics; BBB model capability; PK/PD capabilities; validated drug mechanisms; age relevance; timeline; model limitations; regulatory acceptance history; standalone regulatory evidence flag; MTA required; NGNRI repository
Antibodies	**Host organism, Reactive species, Target antigen, UniProt ID**	Clonality, clone ID, conjugate
Genetic Reagents	**Insert species, Entrez gene ID, Vector backbone, Vector type**	gRNA/shRNA sequence, cloning method, resistance markers, biosafety flag
Computational Tools	**Software type**	Language, version, repo URL, license, container, distribution channel, dependencies; analytical platform support (mass spec, metabolomics, PK/PD modeling…)
Clinical Assessment Tools	**Assessment type, Target population**	NF-specific flag, number of items, psychometric validation, languages, licensing; cognitive/behavioral domains assessed; regulatory acceptance history (FDA/EMA COA endorsement)
Biobanks	**Biobank URL**	Disease types, specimen types, tumor types, preservation methods, access procedure; MTA required; NGNRI repository affiliation
ALL forms	**Tool name + Description + How to acquire**	**Development DOI, Usage DOIs, Synonyms, Developer info, RRID**

New: Translational Decision-Support Metadata

Based on the 2026 NFRTC funder technical specifications, model system forms now capture metadata enabling **go/no-go decisions** — so drug developers, clinical trialists, and regulatory strategists can identify which models answer their specific translational questions.

Drug Testing Capabilities

Added to all 4 model system forms

- **Route of administration**
- **BBB integrity / penetration assessment**
- **PK/PD measurement capabilities**
- **Validated drug mechanisms (MEK, RAS, mTOR…)**
- **Advanced cellular only: BBB model capability flag**
- **Advanced cellular only: standalone regulatory evidence flag**

Model Reliability & Context

Enables convergent evidence evaluation

- **Known model limitations (explicit free text)**
- **Clinical translation history**
- **Regulatory acceptance history (FDA/EMA)**
- **Age relevance (pediatric / adult / both)**
- **Typical timeline to results**
- **Animal model only: induced vs. developmental**

Access & Repository

Reduces bureaucratic friction

- **MTA required? (yes/no)**
- **Deposited in NGNRI repository?**
- **Also on Biobanks: NGNRI affiliation flag**
- **Computational tools: analytical platform support**
- **(mass spec, metabolomics, PK/PD modeling…)**
- **Clinical tools: cognitive/behavioral domains; regulatory COA history**

Also updated: Animal model manifestation enum now includes *Low Grade Glioma* (NF1-LGG) explicitly. NF disorder fields on patient-derived and advanced cellular model forms now include *NF1 Spontaneous Mutation* as an option.

Linking Experimental Findings to Tools

Beyond cataloging tools, the registry captures **how they perform** — connecting specific experimental findings from publications to the tools that generated them. This transforms the registry from a directory into a living evidence base.

1 Select Tool & Observation Type

Dropdowns pre-populated with all existing registry entries by category. Observation types are context-sensitive: a cell line offers drug response, viability, gene expression; an animal model offers tumor growth, survival, behavioral, drug efficacy.

2 Describe the Finding

Free-text description of the specific experimental result or observation, plus the DOI of the paper it comes from.

3 Tag with Ontology (Optional)

Phenotypic observations can be tagged with Mammalian Phenotype (MP) ontology IDs, enabling cross-study comparisons.

4 Attach Files (Optional)

Supporting images, datasets, or figures can be uploaded alongside the observation record.

Observation Types by Category

Tool Type	Example Observation Types
Animal Models	Tumor growth, survival, drug efficacy, behavioral, immune, metabolic
Cell Lines	Drug response, viability, migration/invasion, gene expression, morphological
PDX / Adv. Cellular	Drug response, gene expression, morphological, molecular/biochemical
Antibodies	Validation data, cross-reactivity, usage protocol, quality note
Computational	Performance benchmark, validation result, issue report
Clinical Tools	Psychometric validation, population-specific findings

Why This Matters for the Funding Organization

Portfolio Visibility

Every grant-funded tool is structured, tagged, and searchable — giving program officers a real-time view of what the portfolio has produced, filterable by disease area, model type, or institution.

Maximizing Return on Investment

When tools are registered with acquisition instructions, other funded investigators can access and reuse them immediately — reducing duplication and stretching grant dollars further.

Publication Tracking

The registry links each tool to every paper that has used it. Program managers can pull a citation map showing downstream research productivity enabled by any specific funded tool.

Community Coordination

Biobanks, clinical assessment tools, and patient-derived models are visible across institutions — enabling the program office to actively broker resource sharing and cross-institutional collaborations.

Grant Reporting Support

Grantees can point to registry entries as structured evidence of deliverables met. Program officers can generate standardized tool-output reports directly from the registry.

The submission forms are more than data entry. They are the mechanism by which the organization's investment in tools becomes a permanent, shared, and growing community asset.

WHAT EVERY SUBMISSION CAPTURES

- ✓ Tool identity & scientific description
- ✓ NF disease relevance & model system
- ✓ Publication that first described it
- ✓ All publications that have used it
- ✓ How another lab can access it
- ✓ RRID for durable citation tracking
- ✓ Experimental observations from literature

SUMMARY

A Shared Infrastructure for NF Research

The submission forms are a community-wide standard for capturing, validating, and sharing the experimental tools that drive Neurofibromatosis research.

🔗

Universal Backbone

Name, description, publication DOIs, and acquisition on every form

📦

4 Scientific Groupings

Models · Reagents · Analytical · Infrastructure

✏️

Low Barrier to Entry

Minimal required fields; optional fields enrich over time

🌐

Openly Discoverable

Hosted on Synapse, RRID-indexed, publication-linked